

Improper integrals – 2

Conditional convergence and oscillation

Homework

12 сентября 2026 года

Theoretical minimum. An oscillatory integral converges because of cancellation, and the two standard ways to exhibit it are Bonnet's second mean value theorem (hence the tests of Dirichlet and Abel) and integration by parts, which trades $\int f \sin x$ for $\int f' \cos x$. A phase with φ' monotone and tending to infinity always produces convergence, even when the integrand does not tend to zero. Absolute convergence is a different question and is settled by replacing $|\sin x|$ with its mean value $\frac{2}{\pi}$. The Riemann–Lebesgue lemma holds for every integrable f , with rate $O(1/\lambda)$ for monotone f and no rate in general. Finally, a bound for an oscillatory integral with a stationary phase point loses a factor: $\int e^{i\lambda x^2} \varphi \sim \lambda^{-1/2}$, not λ^{-1} .

1. Prove that $\int_0^\infty \frac{\sin x}{x} dx$ converges, that it is not absolutely convergent, and that the tail satisfies $\int_T^\infty \frac{\sin x}{x} dx = \frac{\cos T}{T} + O\left(\frac{1}{T^2}\right)$.
2. For which $p, q > 0$ does $\int_1^\infty \frac{\sin(x^p)}{x^q} dx$ converge, and for which of them absolutely?
3. Prove that $\int_1^\infty \sin x \sin(x^2) dx$ converges, and decide the same question for $\int_1^\infty \sin x \sin(\ln x) dx$.
4. Construct a continuous $f \geq 0$ on $[0, \infty)$, unbounded, with $\int_0^\infty f$ convergent; then prove that if $\int_0^\infty f$ converges and f is uniformly continuous, then $f(x) \rightarrow 0$.
5. Prove that $\int_1^\infty \frac{\sin x}{x + \sin x} dx$ converges, and explain why the naive argument “the integrand is asymptotically $\frac{\sin x}{x}$, hence the integral converges” is not a proof.

6. (Proposed problems) Prove that $\int_0^\infty \frac{|\sin x|}{1+x^2} dx = \frac{e^2-1}{2e} \ln \frac{e+1}{e-1}$.
7. Let f be non-increasing with $f(x) \rightarrow 0$. Prove that $\int_0^\infty f(x) \sin x dx > 0$ whenever f is not identically zero, and that the same is false for $\int_0^\infty f(x) \cos x dx$.
8. Let f be of bounded variation on $[a, \infty)$ with $f(x) \rightarrow 0$. Prove that $\int_a^\infty f(x) \sin x dx$ converges, and show by an example that “bounded variation” cannot be weakened to “continuous and tending to zero”.
9. Prove that $\int_0^\infty \frac{\sin x}{x} dx$ and $\int_0^\infty \frac{\sin^3 x}{x} dx$ converge and that their values differ by a rational multiple of π ; compute that multiple without computing either integral.
10. Let φ be C^2 on $[a, b]$ with $\varphi' \neq 0$ everywhere. Prove that $|\int_a^b e^{i\lambda\varphi(x)} dx| \leq \frac{C}{\lambda}$ with C depending only on $\min |\varphi'|$ and the variation of φ' , and explain why the same bound fails when φ has a stationary point.